

PROBLEM SET 5—SOLUTIONS

ECON301—ECONOMETRICS I

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions: Due on 20/4/2020 before 12 PM through the Moodle link. You are encouraged to work in groups but each student is required to submit his or her own work. If you worked with a group, please indicated the names of each member in your submission. You can send a computer write-up or scan/picture of your homework. If you scan or take pictures of your homework, please combine the files into a pdf; you can use the following online pdf merger app to do that: <https://combinepdf.com/>

Grading: This assignment is somewhat long. The main purpose of the homework is to let you practice the material, not the grade. Therefore, I will be more lenient with the grading given the circumstances. If you do a well-intentioned effort with the homework, you will get full marks. Try to do as much as you can.

We start by fixing some notation. Consider the $n \times (k + 1)$ matrix of explanatory variables

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}.$$

As in the slides, the $(k + 1) \times 1$ vectors $\mathbf{x}_i = (1, x_{i1}, x_{i2}, \dots, x_{ik})$ collect, for each $i = 1, \dots, n$ the collection of values of the explanatory variables plus the constant for the i th observation in the sample. Thus $\mathbf{x}_i'$ form the *rows* of $\mathbf{X}$. On the other hand, we denote each of the $k + 1$ *columns* of $\mathbf{X}$ as $\mathbf{X}^j$ for $j = 0, \dots, k$. Thus, we can also write the matrix $\mathbf{X}$ as

$$\mathbf{X} = (\mathbf{X}^0, \mathbf{X}^1, \dots, \mathbf{X}^k),$$

note that the column $\mathbf{X}^0$ is a $n \times 1$ vector of ones. It is customary to call this column, the *summer* vector, and to denote it by $\mathbf{s} = (1, \dots, 1)'$. This language will become clear in one of the next exercises.

In what follows, $\mathbf{y} = (y_1, \dots, y_n)'$ is the vector of observations on the dependent variable, $\hat{\mathbf{u}}$ is the vector of OLS residuals, and the matrices $\mathbf{Q}$, $\mathbf{A}$, $\mathbf{N}$, and $\mathbf{M}$ are as in the slides.

1. Show that the OLS estimator $\hat{\boldsymbol{\beta}}$ can be written as

$$\hat{\boldsymbol{\beta}} = \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\sum_{i=1}^n \mathbf{x}_i y_i \right)$$

Dividing each summation by n shows that $\hat{\boldsymbol{\beta}}$ is a function of sample averages.

Solution

Recall that $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. Writing

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix}$$

so that

$$\mathbf{X}' = (\mathbf{x}_1, \dots, \mathbf{x}_n),$$

then we have that

$$\mathbf{X}'\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n) \begin{bmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix} = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i',$$

and

$$\mathbf{X}'\mathbf{y} = (\mathbf{x}_1, \dots, \mathbf{x}_n) \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n \mathbf{x}_i y_i,$$

which proves the expression for $\hat{\boldsymbol{\beta}}$ given in the problem.

2. Let $\hat{\boldsymbol{\beta}}$ be the vector of the OLS estimates.

(a) Show that for any $(k+1) \times 1$ vector $\mathbf{b}$, we can write the sum of squared residuals as

$$\text{SSR}(\mathbf{b}) = \hat{\mathbf{u}}'\hat{\mathbf{u}} + (\hat{\boldsymbol{\beta}} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}).$$

Hint: Write $(\mathbf{y} - \mathbf{Xb})'(\mathbf{y} - \mathbf{Xb}) = (\hat{\mathbf{u}} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}))'(\hat{\mathbf{u}} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}))$ and use the fact that $\mathbf{X}\hat{\mathbf{u}} = \mathbf{0}$.

(b) Explain how the expression for $\text{SSR}(\mathbf{b})$ derived above proves that $\hat{\boldsymbol{\beta}}$ uniquely minimizes $\text{SSR}(\mathbf{b})$ over all possible values of $\mathbf{b}$, assuming that $\mathbf{X}$ is of full rank.

Solution

(a) Using the hint:

$$\begin{aligned} \text{SSR}(\mathbf{b}) &= (\mathbf{y} - \mathbf{Xb})'(\mathbf{y} - \mathbf{Xb}) \\ &= (\hat{\mathbf{u}} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}))'(\hat{\mathbf{u}} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})) \\ &= \hat{\mathbf{u}}'\hat{\mathbf{u}} + \hat{\mathbf{u}}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) + (\hat{\boldsymbol{\beta}} - \mathbf{b})'\mathbf{X}'\hat{\mathbf{u}} + (\hat{\boldsymbol{\beta}} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) \end{aligned}$$

and using the other part of the hint $\mathbf{X}\hat{\mathbf{u}} = \mathbf{0}$ we get that the two middle terms in the previous expression are zero, so that

$$= \hat{\mathbf{u}}'\hat{\mathbf{u}} + (\hat{\boldsymbol{\beta}} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})$$

(b) The term $\hat{\mathbf{u}}'\hat{\mathbf{u}} \geq 0$ since it is a sum of squares. The term $(\hat{\boldsymbol{\beta}} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})$ is also a sum of squares, to see this, note that $\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) = \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{Xb}$ is just a $n \times 1$ vector. Therefore, $(\hat{\boldsymbol{\beta}} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) \geq 0$. Therefore, $\text{SSR}(\mathbf{b})$ is minimized when we put choose $\mathbf{b} = \hat{\boldsymbol{\beta}}$.

3. This exercise is meant to help you get fluent with the computations that appear in the matrix form of the multiple linear regression model. Recall that $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}$, $\hat{\mathbf{y}} = \mathbf{N}\mathbf{y}$ and $\hat{\mathbf{u}} = \mathbf{M}\mathbf{y}$ as in the slides. Show the following as concisely as possible.

(a) $\mathbf{AN} = \mathbf{A}$, $\mathbf{AM} = \mathbf{O}$, $\mathbf{MN} = \mathbf{O}$, $\mathbf{NM} = \mathbf{O}$.

(b) $\mathbf{NX} = \mathbf{X}$, $\mathbf{MX} = \mathbf{O}$.

(c) $\mathbf{N}\hat{\mathbf{y}} = \hat{\mathbf{y}}$, $\mathbf{N}\hat{\mathbf{u}} = \mathbf{0}$.

(d) $\mathbf{M}\hat{\mathbf{y}} = \mathbf{0}$, $\mathbf{M}\hat{\mathbf{u}} = \hat{\mathbf{u}}$.

(e) $\mathbf{X}'\hat{\mathbf{y}} = \mathbf{X}'\mathbf{y}$.

(f) $\mathbf{y}'\hat{\mathbf{y}} = \mathbf{yX}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = \hat{\boldsymbol{\beta}}'\mathbf{Q}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}}'\hat{\mathbf{y}}$.

(g) $\hat{\mathbf{u}}'\hat{\mathbf{u}} = \mathbf{y}'\mathbf{M}\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat{\mathbf{y}}'\hat{\mathbf{y}}$.

Solution

(a) $\mathbf{AN} = \mathbf{AXA} = \mathbf{Q}^{-1}\mathbf{X}'\mathbf{XQ}^{-1}\mathbf{X}' = \mathbf{Q}^{-1}\mathbf{QQ}^{-1}\mathbf{X}' = \mathbf{Q}^{-1}\mathbf{X}' = \mathbf{A}$; $\mathbf{AM} = \mathbf{A}(\mathbf{I} - \mathbf{N}) = \mathbf{AI} - \mathbf{AN} = \mathbf{A} - \mathbf{A} = \mathbf{O}$; $\mathbf{MN} = (\mathbf{I} - \mathbf{N})\mathbf{N} = \mathbf{IN} - \mathbf{NN} = \mathbf{N} - \mathbf{N} = \mathbf{O}$; $\mathbf{NM} = (\mathbf{MN})' = \mathbf{O}$.

(b) $\mathbf{NX} = \mathbf{XAX} = \mathbf{XI} = \mathbf{X}$; $\mathbf{MX} = (\mathbf{I} - \mathbf{N})\mathbf{X} = \mathbf{IX} - \mathbf{NX} = \mathbf{X} - \mathbf{X} = \mathbf{O}$.

(c) $\mathbf{N}\hat{\mathbf{y}} = \mathbf{NNy} = \mathbf{Ny} = \hat{\mathbf{y}}$; $\mathbf{N}\hat{\mathbf{u}} = \mathbf{NM}\mathbf{y} = \mathbf{Oy} = \mathbf{0}$.

(d) $\mathbf{M}\hat{\mathbf{y}} = \mathbf{M}\mathbf{Ny} = \mathbf{Oy} = \mathbf{0}$; $\mathbf{M}\hat{\mathbf{u}} = \mathbf{MM}\mathbf{y} = \mathbf{My} = \hat{\mathbf{u}}$.

(e) $\mathbf{X}'\hat{\mathbf{y}} = \mathbf{X}'\mathbf{Ny} = (\mathbf{NX})'\mathbf{y} = \mathbf{X}'\mathbf{y}$.

$$\begin{aligned}
\text{(f) } \mathbf{y}'\hat{\mathbf{y}} &= \mathbf{y}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y}'\mathbf{X}\mathbf{A}\mathbf{y} = \mathbf{y}'\mathbf{N}\mathbf{y} = \mathbf{y}'\mathbf{N}'\mathbf{y} = \mathbf{y}'\mathbf{A}'\mathbf{X}\mathbf{y} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = \mathbf{y}'\mathbf{X}\mathbf{Q}^{-1}\mathbf{X}'\mathbf{y} = \mathbf{y}'\mathbf{X}\mathbf{Q}^{-1}\mathbf{Q}\mathbf{Q}^{-1}\mathbf{X}'\mathbf{y} = \\
&\mathbf{y}'\mathbf{A}'\mathbf{Q}\mathbf{A}\mathbf{y} = \hat{\boldsymbol{\beta}}'\mathbf{Q}\hat{\boldsymbol{\beta}} = \mathbf{y}'\mathbf{N}\mathbf{y} = \mathbf{y}'\mathbf{N}\mathbf{N}\mathbf{y} = \mathbf{y}'\mathbf{N}'\mathbf{N}\mathbf{y} = \hat{\mathbf{y}}'\hat{\mathbf{y}}. \\
\text{(g) } \hat{\mathbf{u}}'\hat{\mathbf{u}} &= \mathbf{y}'\mathbf{M}'\mathbf{M}\mathbf{y} = \mathbf{y}'\mathbf{M}\mathbf{M}\mathbf{y} = \mathbf{y}'\mathbf{M}\mathbf{y} = \mathbf{y}'(\mathbf{I} - \mathbf{N})\mathbf{y} = \mathbf{y}'\mathbf{I}\mathbf{y} - \mathbf{y}'\mathbf{N}\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat{\mathbf{y}}'\hat{\mathbf{y}}.
\end{aligned}$$

4. Under the standard assumptions MLR1–MLR5, show that $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}$ is uncorrelated with $\hat{\mathbf{u}} = \mathbf{M}\mathbf{y}$.

Solution

$$C(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}) = C(\mathbf{A}\mathbf{y}, \mathbf{M}\mathbf{y}) = \mathbf{A}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{A}\mathbf{M} = \sigma^2\mathbf{O} = \mathbf{O}.$$

5. Let $\hat{\boldsymbol{\beta}}$ be the OLS estimate from the regression of $\mathbf{y}$ on $\mathbf{X}$. Let $\mathbf{Z} = \mathbf{X}\mathbf{T}$ where $\mathbf{X}$ is the matrix of observations of the explanatory variables and has rank $k + 1$, and $\mathbf{T}$ is a $(k + 1) \times (k + 1)$ matrix with rank $k + 1$. Let $\tilde{\boldsymbol{\beta}}$ denote the OLS estimate from a regression of $\mathbf{y}$ on $\mathbf{Z}$, and let $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{u}}$ denote the fitted values and residuals, respectively, of this regression.

- Show that $\tilde{\boldsymbol{\beta}} = \mathbf{T}^{-1}\hat{\boldsymbol{\beta}}$.
- Show that $\tilde{\mathbf{y}} = \hat{\mathbf{y}}$ and conclude that $\hat{\mathbf{u}} = \tilde{\mathbf{u}}$.
- Show that the variance matrix for $\tilde{\boldsymbol{\beta}}$ is $\sigma^2\mathbf{T}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}^{-1'}$.
- Let $\hat{\beta}_j$ be the OLS estimates of regressing y_i on $1, x_{i1}, \dots, x_{ik}$ and let $\tilde{\beta}_j$ be the OLS estimates from the regression of y_i on $1, a_1x_{i1}, \dots, a_kx_{ik}$, where at least one of the $a_i \neq 0$. Use the results from part (a) to find the relationship between $\tilde{\beta}_j$ and $\hat{\beta}_j$. What does this result have to do with our discussion of changing the measurement units of the dependent and explanatory variables?

Solution

- (a) Here I use the properties of inverses in appendix D.

$$\begin{aligned}
\tilde{\boldsymbol{\beta}} &= (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} = (\mathbf{T}'\mathbf{X}'\mathbf{X}\mathbf{T})^{-1}\mathbf{T}'\mathbf{X}'\mathbf{y} = \mathbf{T}^{-1}(\mathbf{T}'\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\mathbf{X}'\mathbf{y} = \mathbf{T}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}^{-1'}\mathbf{T}'\mathbf{y} \\
&= \mathbf{T}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{y} = \mathbf{T}^{-1}\mathbf{A}\mathbf{y} = \mathbf{T}^{-1}\hat{\boldsymbol{\beta}}.
\end{aligned}$$

- (b) Using the previous item:

$$\tilde{\mathbf{y}} = \mathbf{Z}\mathbf{T}^{-1}\hat{\boldsymbol{\beta}} = \mathbf{X}\mathbf{T}\mathbf{T}^{-1}\hat{\boldsymbol{\beta}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}}.$$

Since the fitted values are equal, then the estimated errors are also equal: $\hat{\mathbf{u}} = \tilde{\mathbf{u}}$.

- (c) Recall that $V(\hat{\boldsymbol{\beta}}|\mathbf{X}) = \sigma^2\mathbf{Q}^{-1}$. Thus $V(\tilde{\boldsymbol{\beta}}|\mathbf{X}) = \sigma^2\mathbf{T}^{-1}\mathbf{Q}^{-1}\mathbf{T}^{-1'}$.
- (d) The transformation proposed in this item is equivalent to post-multiplying $\mathbf{X}$ by the following matrix:

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & a_1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & \cdots & a_k \end{bmatrix}$$

$\mathbf{T}$ has the coefficients $(1, a_1, \dots, a_k)$ in the diagonal and zero everywhere else. Note that the inverse of $\mathbf{T}$ is:

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1/a_1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & \cdots & 1/a_k \end{bmatrix}$$

Thus, given the results in the previous items, $\tilde{\boldsymbol{\beta}} = \mathbf{T}^{-1}\hat{\boldsymbol{\beta}} = (1, \hat{\beta}_1/a_1, \dots, \hat{\beta}_k/a_k)'$. Naturally, this specific transformation is just a change in the units of measurement of the variables, and this exercise shows how that change affects the original estimates.

6. This exercise has to do with the section about partitioned regression in the slides. Consider the customary situation, where the regression includes an intercept, and the first column of $\mathbf{X}$ is the summer vector $\mathbf{s} = (1, \dots, 1)'$. Let $\mathbf{M}_s = \mathbf{I} - \mathbf{s}(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'$.

- (a) Show that $\mathbf{M}_s\mathbf{y}$ is the vector of residuals from a regression of $\mathbf{y}$ on the summer vector alone.
- (b) Show that $\mathbf{y}'\mathbf{M}_s\mathbf{y} = \sum_i (y_i - \bar{y})^2$.
- (c) Suppose assumptions MLR1–MLR5 hold. Use the results in the slides on computing expected values of random vectors to show that

$$E \left[\sum_i (y_i - \bar{y})^2 \right] = (n-1)\sigma^2 + \boldsymbol{\beta}'_2 \mathbf{X}_2^* \mathbf{X}_2^* \boldsymbol{\beta}_2,$$

where $\boldsymbol{\beta}_2$ is the $k \times 1$ sub-vector that remains when the first element of $\boldsymbol{\beta}$ is deleted (that is, the population slopes corresponding to the k explanatory variables besides the constant), $\mathbf{X}_2$ is the $n \times k$ sub-matrix that remains when the first column of $\mathbf{X}$ is deleted, and $\mathbf{X}_2^* = \mathbf{M}_s \mathbf{X}_2$.

- (d) Use the results in here to evaluate the claim that I made in page 37 of the slides `OLS-matrixform.pdf`, about the denominator of the adjusted R^2 .

Solution

- (a) The summer vector is $\mathbf{s} = (1, \dots, 1)'$. Then $\mathbf{s}'\mathbf{s} = n$ so that $\mathbf{s}'\mathbf{s}^{-1} = 1/n$. Also, $\mathbf{s}'\mathbf{y} = \sum_{i=1}^n y_i$ so that in a regression of $\mathbf{y}$ on $\mathbf{s}$ alone we would obtain the coefficient $\mathbf{s}'\mathbf{s}^{-1}\mathbf{s}'\mathbf{y} = (1/n) \sum_{i=1}^n y_i = \bar{y}$. The fitted values are obtained by multiplying the summer vector by the coefficient we obtained: $\mathbf{s}\bar{y} = (\bar{y}, \dots, \bar{y})'$. The residuals are $\mathbf{y} - (\bar{y}, \dots, \bar{y})' = (y_1 - \bar{y}, \dots, y_n - \bar{y})'$ which are indeed obtained by $\mathbf{M}_s\mathbf{y} = \mathbf{y} - \mathbf{s}(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}'\mathbf{y}$.
- (b) Note that $\mathbf{M}_s$ is idempotent and symmetric. Thus $(\mathbf{M}_s\mathbf{y})'(\mathbf{M}_s\mathbf{y}) = \mathbf{y}'\mathbf{M}_s\mathbf{y} = (y_1 - \bar{y}, \dots, y_n - \bar{y})(y_1 - \bar{y}, \dots, y_n - \bar{y})' = \sum_{i=1}^n (y_i - \bar{y})^2$.
- (c) Firstly, note that we are partitioning $\mathbf{X} = (\mathbf{s}, \mathbf{X}_2)$, where the first column is the summer vector and the remaining columns are in $\mathbf{X}_2$. Thus, $\mathbf{X}\boldsymbol{\beta} = (\mathbf{s}, \mathbf{X}_2)(\boldsymbol{\beta}_1, \boldsymbol{\beta}_2)' = \mathbf{s}\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2$. I establish some algebraic facts:

$$\mathbf{M}_s = \mathbf{I} - \mathbf{s}(\mathbf{s}'\mathbf{s})^{-1}\mathbf{s}' = \mathbf{I} - \mathbf{s}(1/n)\mathbf{s}' = \mathbf{I} - \begin{bmatrix} 1/n & \dots & 1/n \\ \vdots & \vdots & \vdots \\ 1/n & \dots & 1/n \end{bmatrix} = \begin{bmatrix} 1-1/n & -1/n & \dots & -1/n \\ -1/n & 1-1/n & \dots & -1/n \\ \vdots & \vdots & \ddots & \vdots \\ -1/n & -1/n & \dots & 1-1/n \end{bmatrix},$$

thus, we have that $\text{tr}(\mathbf{M}_s) = \sum_{i=1}^n (1 - 1/n) = n - 1$. We also have that $\mathbf{M}_s\mathbf{X} = \mathbf{M}_s(\mathbf{s}, \mathbf{X}_2) = \mathbf{M}_s\mathbf{s} + \mathbf{M}_s\mathbf{X}_2$, but $\mathbf{M}_s\mathbf{s} = \mathbf{0}$ as can be easily verified, so the previous expression is $\mathbf{M}_s\mathbf{X} = \mathbf{M}_s\mathbf{X}_2$.

Thus, using propositions 6 in the slides:

$$\begin{aligned} E \left[\sum_i (y_i - \bar{y})^2 \right] &= E[\mathbf{y}'\mathbf{M}_s\mathbf{y}] = \text{tr}(\mathbf{M}_s\sigma^2\mathbf{I}) + \boldsymbol{\beta}'\mathbf{X}'\mathbf{M}_s\mathbf{X}\boldsymbol{\beta} \\ &= \sigma^2 \text{tr}(\mathbf{M}) + (\mathbf{s}\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2)'\mathbf{M}_s(\mathbf{s}\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) \\ &= \sigma^2(n-1) + \boldsymbol{\beta}'_2 \mathbf{X}_2' \mathbf{M}_s \mathbf{X}_2 \boldsymbol{\beta}_2 \\ &= \sigma^2(n-1) + \boldsymbol{\beta}'_2 \mathbf{X}_2^* \mathbf{X}_2^* \boldsymbol{\beta}_2 \end{aligned}$$

- (d) The denominator of the $\bar{R}^2$ is $\sum_i (y_i - \bar{y})^2 / (n-1)$. Given the previous results, we have that $E[\sum_i (y_i - \bar{y})^2 / (n-1)] = \sigma^2 + (\boldsymbol{\beta}'_2 \mathbf{X}_2^* \mathbf{X}_2^* \boldsymbol{\beta}_2) / (n-1)$, therefore, the remark in that slide is true: the denominator is not an unbiased estimator of σ^2 .